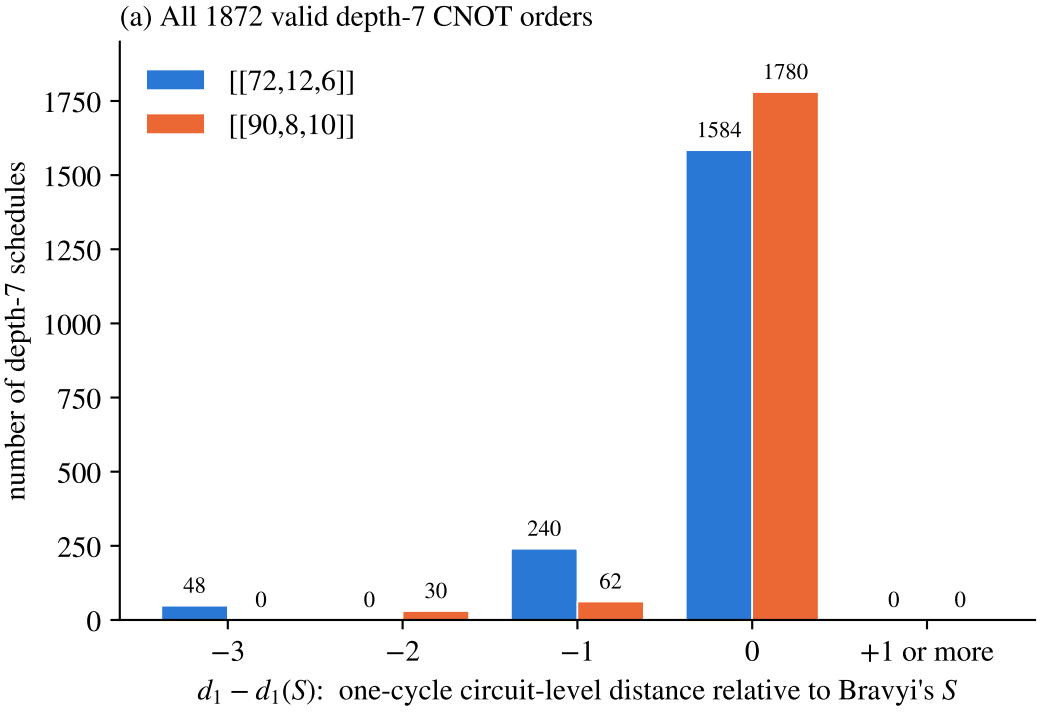
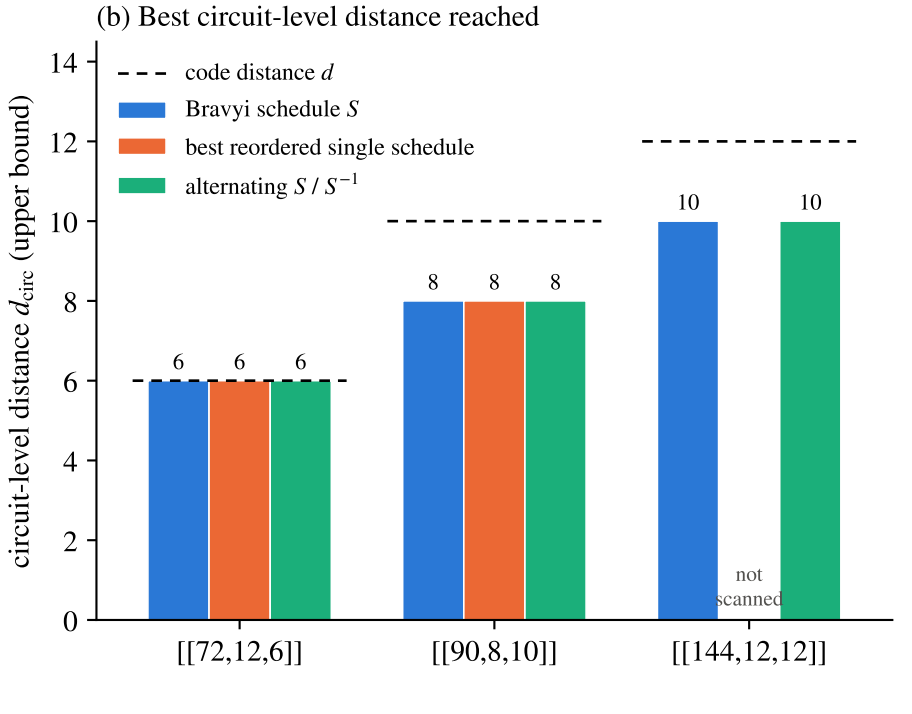

(a) All 1872 valid depth-7 CNOT orders
[[72,12,6]]
[[90,8,10]]
number of depth-7 schedules
1750
1500
1250
1000
750
500
250
0
48
0
0
30
240
62
1584
1780
0
0
−3
−2
−1
0
+1 or more
$d_1 - d_1(S)$: one-cycle circuit-level distance relative to Bravyi's S


(b) Best circuit-level distance reached
code distance d
Bravyi schedule S
best reordered single schedule
alternating S / S^{-1}
circuit-level distance d_{circ} (upper bound)
14
12
10
8
6
4
2
0
6
6
6
8
8
8
10
not scanned
10
[[72,12,6]]
[[90,8,10]]
[[144,12,12]]